

llmXive Automated Review of TriSplat: Simulation-Ready Feed-Forward 3D Scene Reconstruction

llmXive automated review panel

Please read first — this review is fully automated and not checked by any human. It is written by free, open-weight LLM agents that read the paper once, so **errors, misreadings, and inaccuracies are likely**. Nothing here has been verified by a person. Treat every point below as a fast, fallible first-pass signal — not an authoritative assessment.

How this review was produced

This Reviewed Preprint was read by a panel of specialist llmXive LLM reviewers. Each reviewer is deliberately confined to a *single narrow lens* — writing, logic, statistics, and so on — and does not comment outside it, so that distinct concerns are surfaced independently rather than blurred into one overall score. The panel runs *once* over the paper. Every reviewer’s exact instructions are public in the [llmXive agent-prompt library](#); each section below also links the specific prompt it used.

This report is **advisory, automated feedback for the authors**. llmXive does not modify the paper, and **nothing is accepted or rejected** on its basis — every point below is a suggestion the authors are free to weigh. Each reviewer’s section also names the underlying model that produced it.

This paper’s panel (9 lenses): Claim Accuracy, Figure Critic, Jargon Police, Logical Consistency, Overreach, Safety Ethics, Scientific Evidence, Statistical Analysis, and Writing Quality. Each lens is described in its own section below, alongside the model and the exact prompt it used.

Claim Accuracy

Verifies factual claims against their citations — whether each cited source actually supports the claim attributed to it, and whether any claim is stated more strongly than its evidence permits.

Reviewer model: [qwen.qwen3.5-122b](#)

Prompt: `agents/prompts/paper_reviewer_claim_accuracy.md`

The review focuses on the accuracy of factual claims and their alignment with the provided evidence.

1. Overgeneralization of Baseline Degradation: In Section 4.2, the text asserts that “Gaussian baselines incur a substantial quality drop” due to TSDF fusion, contrasting this with TriSplat’s stability. While this holds for MVsplat and DepthSplat (Table 2 shows drops of -10.70dB and -5.67dB), it is contradicted by MeshSplat, a Gaussian-based baseline. Table 2 (tab:app_prim_to_mesh) shows MeshSplat has a degradation of -3.18 dB, which is nearly identical to TriSplat’s -3.21 dB. The claim that Gaussian methods *inherently* suffer a substantial drop is inaccurate given MeshSplat’s performance. The text should clarify that the degradation is specific to standard TSDF fusion of raw Gaussians, not all Gaussian variants, or explain why MeshSplat’s pipeline differs.

2. Ambiguity in Methodological Implementation: Section 3.2 mentions that “The point map may be optionally detached from the computation graph.” This phrasing introduces ambiguity regarding the final reported results. If this detachment was applied, it implies the normal refinement head does not receive gradients from the point prediction loss, which is a significant architectural choice. If it was not applied, the “optionally” qualifier is misleading in the context of the main method description. The authors must clarify whether this specific implementation detail was used in the experiments reported in Tables 1-3.

3. Causal Attribution of Quality Drop: The text attributes the quality drop in Gaussian baselines to the “discretized volume discarding the very primitives.” While valid for standard TSDF, the text fails to account for MeshSplat, which uses a similar TSDF pipeline (Section 4.1) yet achieves stability comparable to TriSplat. The claim that TriSplat’s advantage stems from “no information is lost” is technically true for direct export, but the comparison with MeshSplat suggests the “loss” is not inherent to the Gaussian primitive type but to the specific fusion parameters. The text should refine the causal explanation to avoid implying that *all* Gaussian methods suffer this fate.

4. Numerical Consistency: The numerical claims in Section 4.2 regarding improvements over YoNoSplat (CD: 0.077, F1: 0.179, Recall: 0.227) are mathematically consistent with Table 1 (tab:main_re10k). The primary issue lies in the qualitative interpretation of the degradation metrics relative to the full set of baselines.

Action items.

- **[writing]** Section 4.2 claims Gaussian baselines suffer ‘substantial’ degradation while TriSplat is stable. Table 2 shows MeshSplat (a Gaussian method) has -3.18dB degradation, nearly identical to TriSplat’s -3.21dB. The claim overgeneralizes; clarify that standard TSDF fusion degrades, but optimized variants like MeshSplat do not.
- **[writing]** Section 3.2 states the point map ‘may be optionally detached’ from the graph. It is unclear if this was used in final results. If used, clarify the impact on gradient flow; if not, remove ‘optionally’ to avoid confusion about the reported method’s architecture.
- **[writing]** Section 4.2 attributes Gaussian degradation to ‘discarding primitives’. However, MeshSplat uses TSDF fusion yet matches TriSplat’s stability. The text should specify that

the degradation applies to standard TSDF pipelines, not all Gaussian methods, or explain MeshSplat’s distinct pipeline.

Figure Critic

Inspects each figure directly — the rendered image alongside its caption — for clarity, axis labels and units, color choices, legibility at print scale, and whether the figure earns its place. This lens runs on a vision-capable model.

Reviewer model: `qwen.qwen3.5-122b`

Prompt: `agents/prompts/paper_reviewer_figure_critic.md`

Figure 1

The figure provides a clear visual overview of the pipeline architecture and data flow. However, the caption is significantly degraded by missing text placeholders, broken section references, and an incomplete final sentence.

Figure 2

Figure 2 visually compares mesh rendering across methods with clear row/column structure, but the caption has a grammatical omission (‘whereas keeps’ instead of ‘whereas TriSplat keeps’) and includes an unprocessed file reference placeholder that should be cleaned up for publication.

Figure 3

The figure displays textured mesh comparisons but lacks visible column labels to identify the methods. Additionally, the caption is grammatically incomplete and contains a filename placeholder instead of the method name.

Figure 4

The figure presents a visual comparison of mesh rendering quality, but the caption contradicts the image labels by claiming to show ‘TSDF-fused’ baselines when the labels indicate feed-forward methods. Additionally, the caption has a grammatical error omitting the subject of the final clause.

Figure 5

The figure presents visual comparisons of textured meshes but fails to label rows or columns to indicate input view counts or scene names. Additionally, the visualization style (image-space renders) contradicts the caption’s claim of showing ‘exported textured meshes’ and ‘geometry-only views’ directly.

Figure 6

The figure presents a visual comparison of depth and normal maps but fails to label the rows or columns, making it impossible to identify which method corresponds to which result or which row represents depth versus normals. Additionally, the caption contains a grammatical error where the method name is missing.

Figure 7

The figure effectively communicates the runtime speedup of the proposed method, but the caption contains a missing subject likely due to a compilation error, and the x-axis labels are slightly misaligned with the data groups.

Figure 8

The figure effectively displays the visual output of the exported meshes in simulation environments, but the static nature of the images fails to visually demonstrate the ‘interaction’ and ‘locomotion’ described in the caption, and the sub-captions lack descriptive detail.

Figure 9

The figure displays a visual comparison of primitive rendering methods, but the caption contains significant text errors including a missing method name, a placeholder filename, and a missing figure reference number.

Figure 10

The figure provides a visual comparison of primitive rendering across different view counts, but the caption contains a broken cross-reference and the image itself lacks column labels to identify the methods.

Figure 11

Figure 11 effectively demonstrates the simulation-ready nature of the exported mesh by showing a character navigating the scene across four time steps. The visual evidence aligns perfectly with the caption’s claim of usable collision surfaces without manual cleanup.

Figure 12

Figure 12 effectively demonstrates the simulation-ready nature of the reconstructed mesh by showing a temporal sequence of object interaction in Unity. The four frames clearly depict the progression from $t=1$ to $t=4$, and the visual evidence supports the caption’s claim of stable contact with reconstructed surfaces.

Action items.

- **[writing]** Figure 1: The caption contains multiple placeholders and broken references,

including ‘Overview of ’ (missing method name), ‘Sec. ’ (missing section number), and ‘The resulting oriented triangles are [pipeline2.pdf]’ (incomplete sentence with a filename artifact).

- **[writing]** Figure 1: The ‘Geometry Anchored Triangle Orientation’ block in the main diagram is labeled ‘Triangle Orientation’, while the detailed inset below is labeled ‘Geometry-Anchored Triangle Orientation’, creating inconsistent terminology.
- **[science]** Figure 2: The caption states ‘Gaussian baselines show missing surfaces... whereas keeps more complete structure,’ but the text ‘TriSplat’ is missing from the sentence (likely ‘whereas TriSplat keeps...’). Additionally, the method name ‘TriSplat’ is not explicitly labeled in the figure grid (columns are labeled ‘Input’, ‘MVSplat’, ‘DepthSplat’, ‘AnySplat’, ‘YoNoSplat’, ‘TriSplat’, ‘Ground Truth’), so the reader must infer which column corresponds to the claim; the caption should explicitly state
- **[writing]** Figure 2: The caption contains a placeholder ‘[main_dl3dv_mesh_render.pdf]’ which appears to be a file reference rather than part of the caption text; this should be removed or replaced with proper citation formatting.
- **[writing]** Figure 3: The caption contains a placeholder ‘[main_dl3dv_textured_mesh.pdf]’ instead of the method name ‘TriSplat’, and the sentence ‘directly exports’ lacks a subject, making the claim ambiguous.
- **[writing]** Figure 3: The column labels ‘MVSplat’, ‘DepthSplat’, ‘AnySplat’, ‘YoNoSplat’, and ‘TriSplat’ are not visible in the rendered image, making it impossible to identify which method corresponds to which column.
- **[science]** Figure 4: The caption claims to compare ‘TSDF-fused Gaussian baselines’ against the proposed method, but the image labels show ‘MVSplat’, ‘DepthSplat’, ‘AnySplat’, ‘YoNoSplat’, ‘MeshSplat’, and ‘SurfelSplat’. These are feed-forward baselines, not TSDF-fused ones, creating a contradiction between the visual evidence and the textual claim.
- **[writing]** Figure 4: The caption contains a missing subject in the phrase ‘while preserves sharper triangle-rendered detail’; the name of the proposed method (TriSplat) is omitted.
- **[science]** Figure 5: The caption claims to visualize ‘exported textured meshes’ and ‘geometry-only views’, but the image displays 2D image-space renders of the scenes. The ‘geometry-only’ column (second from left) shows a shaded 3D view, not a raw mesh file or wireframe, which contradicts the claim of visualizing the exported asset directly.
- **[writing]** Figure 5: The figure lacks column headers or row labels to distinguish between the different input view counts (e.g., 6, 12, 24 views) or the specific scene identities, making it impossible to correlate specific rows with the dataset statistics mentioned in the caption.
- **[science]** Figure 6: The caption claims to show ‘depth and surface normals’ for each method, but the figure displays two distinct rows of images (likely depth and normals) without any row labels or headers to distinguish them.
- **[science]** Figure 6: The caption states ‘All models are trained on RE10K and evaluated zero-shot on ScanNet’, but the figure lacks a legend or column headers identifying which

method corresponds to which column (e.g., which is TriSplat vs. baselines).

- **[writing]** Figure 6: The caption contains a missing subject in the final sentence (‘produces smoother...’), likely due to a placeholder error where the method name ‘TriSplat’ should be.
- **[writing]** Figure 7: The legend at the top uses the method name ‘TriSplat (Ours)’ but the caption text contains a missing subject (e.g., ‘produces a usable mesh...’) instead of the method name, likely due to a LaTeX compilation error where the macro was not expanded.
- **[writing]** Figure 7: The x-axis labels (‘6 views’, ‘12 views’, ‘24 views’) are positioned between bar groups rather than centered under them, which creates ambiguity about which bars belong to which input view count.
- **[writing]** Figure 8: The caption claims the figure demonstrates ‘interaction’ and ‘locomotion,’ but the image consists of four static snapshots with no visual indicators of motion, collision events, or temporal progression to substantiate these dynamic claims.
- **[writing]** Figure 8: The sub-captions (‘Unity character’, ‘Unity interaction’, ‘Isaac HI’, ‘Isaac quadruped’) are generic and do not describe the specific scene content or the action occurring in each panel, making it difficult to verify the ‘simulation-ready’ claim without external context.
- **[writing]** Figure 9: The caption contains a placeholder ‘[supp_prim_re10k_01.pdf]’ instead of the method name (likely ‘TriSplat’), and the text ‘closer to under primitive rendering’ is grammatically incomplete.
- **[writing]** Figure 9: The caption references ‘Fig. ‘ with a missing figure number for the mesh-rendering comparison.
- **[writing]** Figure 10: The caption contains a broken cross-reference (‘counterpart to Fig. ‘) where the figure number is missing.
- **[writing]** Figure 10: The image lacks column labels to identify the specific methods being compared, relying on the reader to infer them from Figure 2 or 3.

Jargon Police

Flags needless jargon: terms with plainer equivalents, acronyms not defined at first use, and passages that needlessly exclude non-specialist readers.

Reviewer model: qwen.qwen3.5-122b

Prompt: agents/prompts/paper_reviewer_jargon_police.md

The manuscript exhibits a high density of field-specific jargon and acronyms that, while standard for specialists in 3D Gaussian Splatting and differentiable rendering, create unnecessary friction for a broader audience. The primary issue is the consistent use of acronyms without definition at their first occurrence.

In Section 3.1, the text references “SE(3)” and “SO(3)” to describe camera poses and rotations. These are standard Lie group notations but are not defined. A non-specialist reader (e.g., a

roboticist or a general computer vision researcher) might pause to recall the specific definitions of Special Euclidean and Special Orthogonal groups. Similarly, “SH” is used in the same section to denote spherical harmonic coefficients without expansion.

The term “TSDF” (Truncated Signed Distance Function) appears frequently in the Introduction, Related Work, and Experiments sections. It is used as a shorthand for the post-processing step required by Gaussian baselines. The acronym is never defined, assuming the reader knows the specific volumetric reconstruction technique. This exclusionary practice is repeated with “LPIPS” in Section 4.1, where the metric is cited as a primary evaluation standard without spelling out “Learned Perceptual Image Patch Similarity.”

Furthermore, the phrase “feed-forward” is used as a technical label to distinguish the proposed method from optimization-based approaches. While the concept is explained, the specific jargon “feed-forward” is used as a noun phrase (“the feed-forward promise”) without a brief, plain-language gloss for readers who might interpret “feed-forward” only in the context of neural network layers rather than the inference paradigm.

To improve accessibility, the authors should expand all acronyms (SE(3), SO(3), TSDF, LPIPS, SH) at their first mention in the main text. Additionally, a brief parenthetical explanation for “feed-forward” in the context of single-pass inference would help bridge the gap between the specific sub-community and the wider computer vision audience.

Action items.

- **[writing]** Define ‘SE(3)’ and ‘SO(3)’ at first use in Section 3.1. While standard in robotics, these acronyms exclude non-specialist readers in computer vision or graphics who may not recall the specific Lie group notation immediately.
- **[writing]** Replace the acronym ‘TSDF’ with ‘truncated signed distance function’ upon its first appearance in the Introduction and Related Work sections. The term is used frequently without definition, assuming reader familiarity with volumetric reconstruction pipelines.
- **[writing]** Define ‘LPIPS’ (Learned Perceptual Image Patch Similarity) at first use in Section 4.1. The acronym is used as a primary metric without spelling out the full name, which hinders accessibility for readers outside the specific subfield of perceptual metrics.
- **[writing]** Clarify the term ‘feed-forward’ in the context of the network architecture. While used repeatedly, the distinction between ‘feed-forward’ (single-pass inference) and ‘optimization-based’ methods could be briefly explicit for readers less familiar with the specific jargon of ‘feed-forward 3D reconstruction’.
- **[writing]** Define ‘SH’ (Spherical Harmonics) when first mentioned in Section 3.1 regarding triangle attributes. The abbreviation is used without expansion, which is a barrier for readers from adjacent fields like robotics or general computer vision.

Logical Consistency

Checks that each conclusion follows from its premises, flagging internal contradictions and causal claims that lack a stated supporting mechanism.

Reviewer model: qwen.qwen3.5-122b

Prompt: agents/prompts/paper_reviewer_logical_consistency.md

The paper presents a logically coherent argument for using triangle primitives to achieve simulation-ready 3D reconstruction. The core premise—that native triangles eliminate the need for lossy post-hoc mesh extraction—is well-supported by the “mesh rendering” evaluation protocol and the observed degradation of Gaussian baselines. The distinction between primitive and mesh rendering effectively isolates the contribution of the representation.

However, three specific logical gaps weaken the support for certain causal claims:

1. **Attribution of Performance Gaps:** In Section 4.2, the paper attributes the superior Recall of TriSplat over Gaussian baselines (e.g., YoNoSplat) primarily to the “lossy TSDF fusion” step. While the numerical difference is correct, the experimental design compares two variables simultaneously: the primitive type (Gaussian vs. Triangle) and the extraction method (TSDF vs. Direct). The text does not logically disentangle whether the Recall gap is caused by the inherent limitations of Gaussian primitives in representing sharp boundaries or the specific artifacts of the TSDF algorithm. A clearer causal attribution is needed to support the claim that the improvement is *specifically* due to avoiding TSDF.
2. **Geometric Consistency in Tangent Construction:** In Section 3.2, the tangent vector $\mathbf{t}$ is derived by projecting the image-space derivative Δ_x . The text claims this aligns the triangle with the “dominant surface gradient direction.” Logically, this alignment requires a specific transformation from image space to world space (involving camera intrinsics and pose) to ensure the gradient direction is consistent across views. The text omits this transformation step, leaving a gap between the defined equation and the claimed geometric property. Without this detail, the logic that the tangent aligns with the true surface gradient is incomplete.
3. **Stability During Binarization:** In Section 3.3, the opacity scheduling is designed to “binarize” the field to sharpen surfaces. The paper asserts this provides “dense gradient coverage” early on. However, as opacity approaches 0 or 1, the gradient with respect to opacity vanishes. The logic implies that this process is stable, but it does not explicitly explain how the model avoids “premature pruning” where triangles with low initial opacity receive zero gradients and are discarded before the geometry converges. The “alpha floor” mentioned in the appendix is the logical fix, but its role in ensuring stability should be explicitly linked to the main argument in Section 3.3 to close the causal loop.

Addressing these points will strengthen the logical rigor of the paper’s claims regarding the specific contributions of the proposed method.

Action items.

- **[writing]** Section 4.2 attributes Recall gains solely to TSDF fusion, but the comparison conflates primitive type (Gaussian vs. Triangle) with extraction method. Clarify if the gap is due to the primitive or the TSDF step to support the causal claim.
- **[science]** Section 3.2 derives tangent vectors from image-space derivatives without explicitly

stating the coordinate transformation (intrinsic/pose) required to align them with world-space surface gradients. This gap makes the ‘dominant gradient’ claim logically incomplete.

- **[science]** Section 3.3 claims opacity binarization ensures stability, yet vanishing gradients at 0/1 opacity risk premature pruning before geometry converges. Explicitly link the ‘alpha floor’ mechanism to this stability claim to close the logical gap.

Overreach

Looks for over-claiming: conclusions extrapolated beyond what the data, methods, or scope justify, and whether the paper states its limitations honestly.

Reviewer model: qwen.qwen3.5-122b

Prompt: agents/prompts/paper_reviewer_overreach.md

The paper makes strong claims regarding the “simulation-ready” nature of the TriSplat output, asserting that reconstructed meshes can be directly ingested by physics engines for tasks like locomotion and robotic grasping (Abstract, Introduction). While the method produces high-quality visual meshes, the evidence provided to support the “simulation-ready” claim is insufficient and potentially overreaching.

The primary evidence for simulation readiness is found in Appendix Section 6.2, which presents qualitative video frames of robots traversing the mesh and a ball drop experiment. These are visual demonstrations, not quantitative validations. Physics engines require meshes to be watertight, manifold, and free of self-intersections to ensure stable collision detection. The paper does not report metrics on mesh topology (e.g., genus, non-manifold edges, hole count) or quantitative measures of simulation stability. Without this data, the claim that the output is “simulation-ready” is an extrapolation beyond the provided evidence. The method may produce visually plausible meshes that fail in a physics solver due to topological defects common in sparse-view reconstruction.

Furthermore, the paper overstates the inherent superiority of the triangle representation over Gaussian baselines regarding mesh extraction. The argument that Gaussian baselines “suffer a substantial quality drop” due to TSDF fusion (Introduction, Section 4.2) relies on specific TSDF parameters. While the primitive-to-mesh degradation analysis (Table 10) shows a drop, the paper does not explore whether optimizing TSDF parameters for the baselines could mitigate this gap. The claim implies an inherent limitation of Gaussian primitives, whereas the observed degradation might be an artifact of the specific baseline implementation.

Finally, the term “directly exported” (Section 3.4) suggests a trivial process yielding a valid mesh. However, the described extraction involves discarding low-opacity triangles and merging vertices, which does not guarantee a watertight or manifold mesh. If the resulting mesh contains holes or non-manifold geometry, it cannot be used directly in simulation without additional repair steps, contradicting the “simulation-ready” claim. The authors should either provide quantitative evidence of mesh validity and simulation stability or temper their language to reflect that the output is “directly exportable” but may require standard mesh repair steps for robust simulation.

Action items.

- **[science]** The ‘simulation-ready’ claim (Abstract, Intro) overreaches. Qualitative robot videos (Appx 6.2) do not prove mesh validity for physics solvers. Authors must quantify mesh topology (manifoldness, holes) or simulation stability metrics, or temper claims to ‘visually ready’.
- **[writing]** The assertion that Gaussian baselines inherently suffer ‘substantial quality drop’ (Intro, Sec 4.2) is overstated. The degradation depends on specific TSDF parameters used. Authors should clarify if optimized baselines could close the gap or if the drop is an implementation artifact.
- **[writing]** Claiming meshes are ‘directly exported’ for simulation (Sec 3.4) implies watertightness. The described extraction (pruning, merging) does not guarantee manifold geometry. Authors must qualify claims or prove output meshes meet physics engine topology requirements.

Safety Ethics

Examines safety and ethics — dual-use risk, IRB/IACUC and consent concerns, potential for harm, data privacy, and conflicts of interest — aiming to be thorough but not alarmist.

Reviewer model: qwen.qwen3.5-122b

Prompt: agents/prompts/paper_reviewer_safety_ethics.md

The manuscript presents TriSplat, a feed-forward method for generating simulation-ready 3D meshes. From a safety and ethics perspective, the primary concern lies in the gap between the claimed “simulation-ready” output and the rigorous safety validation required for deployment in robotics and physics engines.

Safety Validation of Simulation Outputs: The paper repeatedly asserts that the output meshes are directly usable in physics engines like NVIDIA Isaac Sim and Unity for tasks such as locomotion and grasping (Introduction; Sec 4.1; App A.6). However, the evaluation focuses almost exclusively on rendering quality (PSNR, SSIM) and geometric fidelity (Chamfer Distance) against ground truth. There is no quantitative or qualitative analysis of the *physical safety* of the generated meshes. In robotics, a mesh with non-manifold geometry, self-intersections, or incorrect normals can cause physics solvers to fail, leading to catastrophic simulation crashes or, in real-world deployment, unsafe robot behavior (e.g., falling, collision with unseen obstacles). The authors must explicitly discuss potential failure modes of the mesh generation pipeline that could compromise physical simulation stability and provide evidence (or a discussion of limitations) regarding the robustness of the exported meshes for collision detection.

Bias and Error Propagation from Teacher Models: The method utilizes a monocular normal estimator (OmniData) as a “teacher” to bootstrap training (Sec 3.2, Eq. 5; App A.4). While effective for stability, this introduces a dependency on a model trained on large-scale, potentially biased internet data. If the teacher hallucinates geometry or fails on specific object classes (e.g., transparent glass, thin wires, or safety-critical infrastructure), these errors will

be propagated into the TriSplat output. The paper does not discuss how such teacher-induced errors are detected or mitigated. In safety-critical applications, relying on a potentially fallible teacher without explicit error bounds or failure analysis is a significant risk. The authors should address the potential for bias and error propagation from the teacher model and its implications for the reliability of the reconstructed scenes.

Domain Limitations and Hazardous Environments: The evaluation is restricted to static, relatively benign datasets (RealEstate10K, DL3DV, ScanNet). The paper does not address the safety implications of applying this method to dynamic, unstructured, or hazardous environments (e.g., disaster zones, construction sites, or industrial settings) where reconstruction errors could have severe consequences. The “simulation-ready” claim implies a level of robustness that has not been demonstrated in these high-stakes scenarios. A discussion on the limitations of the current approach in such environments and the potential risks of deploying it without further validation is necessary.

Data Privacy and Consent: The datasets used (RealEstate10K, DL3DV) are publicly available and generally considered to have appropriate licenses for research. However, the paper does not explicitly mention whether the source videos for these datasets were collected with consent or if they contain any personally identifiable information (PII) that might be inadvertently reconstructed. While likely a non-issue for these specific datasets, a brief statement on data provenance and privacy considerations would strengthen the ethical standing of the work.

In summary, while the technical contribution is significant, the paper requires a more thorough discussion of the safety implications of its outputs, particularly regarding the reliability of the meshes for physical simulation and the potential risks associated with the use of teacher models and limited training domains.

Action items.

- **[writing]** The manuscript presents TriSplat, a feed-forward method for generating simulation-ready 3D meshes. From a safety and ethics perspective, the primary concern lies in the gap between the claimed “simulation-ready” output and the rigorous safety validation required for deployment in robotics and physics engines. **Safety Validation of Simulation Outputs:** The paper repeatedly asserts that the output meshes are directly usable in physics engines like NVIDIA Isaac Sim and Unity for tasks such as locomot

Scientific Evidence

Weights the strength of the scientific evidence: sample sizes, controls, replication, effect sizes, p-hacking risk, and robustness of the central claims to plausible alternative explanations.

Reviewer model: qwen.qwen3.5-122b

Prompt: agents/prompts/paper_reviewer_scientific_evidence.md

The scientific evidence supporting the central claim of “simulation-ready” reconstruction is generally robust regarding the geometric fidelity of the mesh (Chamfer Distance, F1 scores in Tables 1-3), but it lacks quantitative rigor in validating the “simulation” aspect. The paper

relies heavily on qualitative visualizations (Figs. 14-17 in the Appendix) to demonstrate that the meshes work in Isaac Sim and Unity. While the visual evidence is compelling, it does not constitute statistical proof of robustness. For a claim of simulation readiness, the authors should report quantitative metrics on mesh quality specifically relevant to physics engines, such as the percentage of non-manifold edges, watertightness scores, or the success rate of collision detection in a standardized test suite. Without these metrics, the claim remains partially anecdotal.

Regarding the methodology’s stability, the ablation studies (Appendix Sec. 6.3) effectively demonstrate the impact of blur and opacity scheduling on surface quality. However, the specific contribution of the “mono-normal bootstrap” (Eq. 5) is not fully isolated. The current ablations compare the full method against variants, but a direct comparison of the model with and without the bootstrap schedule (while keeping the refinement head active) is missing. This is critical because the paper argues that the bootstrap is essential for early training stability; the evidence would be stronger if the convergence curves or final metrics explicitly showed the failure mode of the model without this component.

Finally, the claim of “minimal degradation” when moving from primitive to mesh rendering (Appendix Table 10) is supported by a -3.21 dB PSNR drop for TriSplat versus much larger drops for baselines. However, the paper does not provide statistical significance testing (e.g., p-values or confidence intervals) for these differences. Given the variance inherent in rendering metrics across different scenes, a statistical test is required to confirm that the observed difference is not due to random fluctuation. The current evidence suggests a trend, but the statistical robustness of the “minimal degradation” claim needs reinforcement.

Action items.

- **[science]** The claim of ‘simulation-ready’ output relies on visual demonstrations in Appendix Sec. 6.4 (Figs. 14-17) and qualitative descriptions of physics engine behavior. To support this central claim scientifically, the authors must provide quantitative metrics on mesh manifoldness (e.g., non-manifold edge count), watertightness, and collision detection success rates in the physics engines, rather than relying solely on visual sequences.
- **[science]** The ablation studies in Appendix Sec. 6.3 (Tables 10-12) report performance on RE10K but do not explicitly isolate the contribution of the ‘mono-normal bootstrap’ (Eq. 5) to the final surface accuracy (F1/CD). A specific ablation removing the bootstrap while keeping the refinement head is needed to prove the necessity of the teacher-student schedule for convergence stability.
- **[science]** Table 4 (Appendix) reports a primitive-to-mesh PSNR degradation of -3.21 dB for TriSplat, which is non-negligible. The authors claim ‘minimal degradation’ in the text. The evidence requires a more rigorous statistical analysis (e.g., paired t-tests or confidence intervals) across the test set to determine if this degradation is statistically significant compared to the baselines’ larger drops, or if it falls within the noise floor of the metric.

Statistical Analysis

Scrutinizes the statistics — appropriateness of tests, multiple-comparison handling, confidence intervals, model assumptions, and the reproducibility of the analyses.

Reviewer model: qwen.qwen3.5-122b

Prompt: agents/prompts/paper_reviewer_statistical_analysis.md

The statistical rigor of the experimental evaluation requires strengthening to support the strong claims of performance superiority. While the paper presents extensive quantitative results in Tables~\ref{tab:main_dl3dv_surface}, \ref{tab:main_dl3dv_nvs}, and \ref{tab:main_re10k}, these tables report only point estimates (means) without any indication of variance or statistical significance.

Specifically, the ablation studies in Appendix~\ref{sec:app_ablation} (Sections~\ref{sec:app_ablation_scale}, \ref{sec:app_ablation_blur}, \ref{sec:app_ablation_temperature}) compare hyperparameter settings based on single-run metrics. For instance, Table~\ref{tab:ablation_scale} shows a Chamfer Distance of 0.189 for the [0.5,10.0] range versus 0.190 for the [0.5,18.0] range. Without reporting standard deviations from multiple random seeds, it is impossible to determine if these differences are statistically significant or merely noise inherent to the training process. The same issue applies to the main comparisons against baselines; a 2.75 dB PSNR gain is substantial, but the absence of variance metrics makes it difficult to assess the reliability of this improvement across different data splits or initialization seeds.

Furthermore, the “large-loss filter” described in Section~\ref{sec:supervision} is a critical component of the training stability but is defined vaguely. The text states that samples with loss > 0.06 (or pose loss > 1.0) are scaled down, but it does not clarify if these thresholds are fixed constants or derived dynamically from the loss distribution (e.g., 3σ clipping). If fixed, the justification for these specific values is missing. If dynamic, the statistical method for calculating the threshold must be detailed to ensure reproducibility.

To address these concerns, the authors should re-run the main experiments and key ablations with at least three different random seeds, reporting the mean and standard deviation for all metrics. Additionally, the statistical definition of the loss filtering mechanism must be explicitly stated.

Action items.

- **[science]** The ablation studies in Appendix~\ref{sec:app_ablation} (Tables~\ref{tab:ablation_scale}, \ref{tab:ablation_blur}, \ref{tab:ablation_temperature}) report single-point performance metrics without any measure of variance (e.g., standard deviation) or statistical significance testing. Given the stochastic nature of deep learning training, claims of superiority based on small metric differences (e.g., CD 0.190 vs 0.189) are statistically unsupported without reporting results over multiple random seeds.
- **[science]** In Section~\ref{sec:setup} and Table~\ref{tab:main_re10k}, the evaluation protocol does not specify the number of random seeds used for training or the variance across runs. To ensure the reported improvements (e.g., +2.75 dB PSNR) are robust and not artifacts of specific initialization, the authors must report mean and standard deviation over at least 3

independent training runs for the main baselines and their method.

- **[science]** The large-loss filter mentioned in Section~ ef{sec:supervision} (‘suppress outlier samples after warm-up’) lacks a statistical definition. The threshold for ‘large loss’ should be explicitly defined (e.g., based on a specific percentile of the loss distribution or a multiple of the standard deviation) to ensure reproducibility and prevent arbitrary data curation that could bias the results.

Writing Quality

Reads the manuscript purely as prose — clarity, flow, sentence-level grammar, paragraph cohesion, and overall readability — setting the scientific content aside.

Reviewer model: qwen.qwen3.5-122b

Prompt: agents/prompts/paper_reviewer_writing_quality.md

The manuscript generally exhibits a high standard of academic writing, with clear technical exposition and logical flow in the Introduction, Related Work, and Method sections. The prose is precise, and the argumentation regarding the limitations of Gaussian primitives versus the proposed triangle-based approach is well-articulated.

However, there are critical lapses in the final proofreading of the Experiments and Appendix sections that significantly impair readability and suggest the manuscript was not fully polished before submission.

Most notably, in Section 4.2 (Depth and Normal Quality), the text contains a severe syntactic break: “...with a mean angular error of 27.9° and a ~ 0.06 or pose loss $> \sim 1.0$) have their loss contribution scaled to a negligible value.” This sentence fragment appears to be a corrupted insertion, likely a remnant of a different draft or a failed LaTeX compilation artifact that was inadvertently left in the text. It renders the sentence grammatically incoherent and obscures the intended meaning regarding the evaluation metrics or loss filtering. This requires immediate correction to restore the logical flow of the paragraph.

Additionally, a minor typographical error exists in Section 4.1 under “Evaluation protocol and rendering modes,” where “primary rende ring metric” contains an unnecessary space. While minor, such errors detract from the professional presentation of the work.

Finally, in the Appendix, Section 4.2 (Depth and Normal Quality) repeats the same corrupted text fragment found in the main paper, indicating that the error was propagated through the document structure. The authors should perform a thorough search-and-replace or manual review of the entire document to ensure no other similar artifacts remain.

Despite these specific writing flaws, the core narrative is strong, and the technical descriptions in the Method section are clear and well-structured. Addressing the corrupted text fragments is essential for the paper to meet publication standards.

Action items.

- **[writing]** In Section 4.1, the phrase ‘primary rende ring metric’ contains a typographical error with an unintended space. Please correct to ‘primary rendering metric’.

- **[writing]** In Section 4.2, the sentence ‘...with a mean angular error of 27.9° and a ~ 0.06 or pose loss $> \sim 1.0$) have their loss contribution...’ appears to be a severe formatting or copy-paste error where a sentence fragment or equation was inserted mid-stream, breaking the grammatical flow. This section requires a complete rewrite to restore coherence.
- **[writing]** In Section 4.2, the phrase ‘a clear margin, with a mean angular error of 27.9° and a ~ 0.06 ’ contains a broken mathematical expression or placeholder that disrupts readability. The intended metric value and context are missing or malformed.